AMRITA KAUR BEDI

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EDUCATION

University of California, Irvine **Expected June 2028**
Bachelor of Science in Biomedical Engineering | Minor in Bioinformatics | Minor in Innovation & Entrepreneurship
Cumulative GPA: 3.96 | Dean's Honor List – **6** Quarters

INDUSTRY EXPERIENCE

Applied Medical **June 2026 – September 2026**
Quality Engineering Intern

- Led 2 cross-functional CAPA investigations under **ISO 13485** and **21 CFR Part 820** using root cause analysis and process evaluation; developed an I-MR chart of failure frequency to assess process behavior and support closure of one CAPA as non-systemic, while recommending corrective actions for the second after identifying a procedural gap
- Redesigned a package peel-test fixture in SolidWorks using human factors principles, incorporating thumb-driven loading to better replicate real-world package opening; created engineering drawings and tested an initial prototype
- Supported the Clinical Development team by consolidating and verifying **70+** competitor claims for energy devices across manufacturer literature and clinical and technical sources, organizing findings into a structured comparison

RESEARCH & PROJECTS

Laparoscopic Trainer Lab Module Development **January 2026 – June 2026**
Project Manager | UCI-Applied Medical Collaboration

- Developed a laparoscopic-training laboratory module for an ABET-accredited biomedical engineering course to teach students how device performance is evaluated against standards through engineering testing
- Designed and executed Arduino-based benchtop testing against **ISO 7741** requirements and developed SOPs, instructional materials, and MATLAB analyses to support course implementation
- Piloted the module with **20+** undergraduate students, tracking task-completion time and collecting survey feedback to identify usability issues and refine the lab experience before course implementation
- Co-authored a published **ASEE** conference paper documenting the lab module and educational implementation

UltraScan Pro: Arduino-Based Ultrasonic Scanning System **September 2025 – December 2025**
Project Manager

- Led a 4-member team in designing and prototyping an ultrasonic scanning system that generated a 180° spatial map of surrounding objects in MATLAB; coordinated integration of Arduino hardware and MATLAB visualization
- Programmed the LCD interface and LED/motion-alert system to translate sensor measurements into real-time feedback
- Placed **2nd out of 19 teams** in the final engineering design competition

Novel Suture Project **May 2025 – Present**
Fabrication and Testing Lead

- Developed a low-cost pressure-sensing system for testing a novel suture technique addressing postpartum hemorrhage
- Reduced electronics platform cost by ~**75%** by replacing commercial quick-start boards with a reusable PCB-based design; soldered and tested **3** PCB iterations, evaluating measurement accuracy, repeatability, and signal quality during hardware validation; calibrated the sensor across a **0-183 kPa** range with $\mathbf{R^2 = 0.997}$
- Selected through competitive abstract review to present the project to **30+** attendees at 2025 National BMES Conference

Downing Lab at UCI **April 2025 – Present**
Data Analytics Lead

- Investigated how CSRP1 influences cytoskeletal organization and focal-adhesion architecture in lung cancer cells to better understand mechanisms of cell migration and their role in cancer progression
- Quantified **20+** morphology parameters in H1299 and A549 cells using ImageJ, Python, and R, finding disruption of actin organization following CSRP1 knockdown **($p < 0.0001$)**, supporting a role for CSRP1 in cytoskeletal organization
- Built a Python image-analysis pipeline to analyze **600+** cell images, revealing adhesion changes after CSRP1 knockdown
- Selected through competitive abstract review to present findings to **30+** attendees at 2025 National BMES Conference

LEADERSHIP & INVOLVEMENT

Biomedical Engineering Society at UCI **January 2025 – Present**
Project Coordinator

- Co-founded and led the MedTech Founders Program to give students a structured path from medical-device idea to prototype, mentoring **20+** participants with **3** teams advancing to the semifinals of UCI entrepreneurship competitions
- Built the program's workshop, mentorship, and design-review structure from the ground up and organized **6+** SolidWorks, Arduino, and programming workshops reaching **50+** undergraduates
- Authored grant applications and funding proposals securing **$2,435** for workshops and student projects

SKILLS

Design & Prototyping: SolidWorks (CSWA Certified), Engineering Drawings, 3D Printing, Laser Cutting, Sensor Integration
Programming & Data Analysis: Python, MATLAB, R, Arduino, ImageJ, SQL, Statistical Analysis, Data Visualization
Quality & Product Development: Benchtop Testing, Statistical Process Control (SPC), CAPA, Root Cause Analysis, ISO 13485